Arizona State University
School of Electrical, Computer and Energy Engineering

EEE 465/591: Photovoltaics Energy Conversion
Practice Problem Set #3 - Solutions

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September 16, 2025

1. E(k) Diagram Classification

Label the three E(k) diagrams as direct or indirect semiconductor:

> **Graph A:** Indirect semiconductor
>
> **Graph B:** Indirect semiconductor
>
> **Graph C:** Direct semiconductor

Note: *Direct semiconductors have the conduction band minimum and valence band maximum at the same k-value (typically k=0), while indirect semiconductors have them at different k-values.*

2. Semiconductor Carrier Concentrations in Silicon

Given: $n_i = 1 \times 10^{10}$ cm^{-3}, $E_G = 1.12$ eV, $E_i = 0.56$ eV (at 300K), kT = 0.0259 eV

Formulas:

$$n = n_i\, e^{(E_F - E_i)/kT}, \quad p = n_i^{\,2}/n$$

$$\textit{p-type: } N_A \approx p, \quad \textit{n-type: } N_D \approx n$$

$$E_F = E_i + kT\,ln(N_D/n_i) \quad (n\text{-}type)$$
$$E_F = E_i - kT\,ln(N_A/n_i) \quad (p\text{-}type)$$

Completed Table:

N_D or N_A	E_F (eV, above E_V)	Majority Carrier Concentration	Minority Carrier Concentration
$N_D = 2{\times}10^{17}$	**0.995 eV**	$n \approx 2.0{\times}10^{17}$ cm^{-3} (e^-)	$p \approx 5.0{\times}10^{2}$ cm^{-3} (h^+)
$\mathbf{N_D \approx 7.5{\times}10^{16}}$	0.97 eV	$n \approx 7.50{\times}10^{16}$ cm^{-3} (e^-)	$p \approx 1.33{\times}10^{3}$ cm^{-3} (h^+)
$\mathbf{N_A \approx 1.58{\times}10^{15}}$	0.25 eV	$p \approx 1.58{\times}10^{15}$ cm^{-3} (h^+)	$n \approx 6.34{\times}10^{4}$ cm^{-3} (e^-)
$N_A = 5{\times}10^{18}$	**0.041 eV**	$p \approx 5.0{\times}10^{18}$ cm^{-3} (h^+)	$n \approx 2.0{\times}10^{1}$ cm^{-3} (e^-)
$\mathbf{N_D \approx 4.2{\times}10^{15}}$	**0.895 eV**	$n = 4.2{\times}10^{15}$ cm^{-3} (e^-)	$p \approx 2.38{\times}10^{4}$ cm^{-3} (h^+)
$\mathbf{N_A \approx 1.0{\times}10^{18}}$	**0.083 eV**	$p \approx 1.0{\times}10^{18}$ cm^{-3} (h^+)	$n = 100$ cm^{-3} (e^-)

Working (step-by-step):

> **(a) $N_D = 2{\times}10^{17}$ cm^{-3} (n-type)**
>
> $n \approx N_D = 2{\times}10^{17}$
>
> $p = n_i^2/n = 10^{20}/(2{\times}10^{17}) \approx 5.0{\times}10^2$
>
> $E_F = E_i + kT\ ln(N_D/n_i) = 0.56 + 0.0259\ ln(2{\times}10^7) \approx$ **0.995 eV**

> **(b) $E_F = 0.97$ eV ($> E_i$) ⇒ n-type**
>
> $n = n_i\, e^{(E_F - E_i)/kT} = 10^{10}\, e^{(0.97-0.56)/0.0259} \approx \mathbf{7.50{\times}10^{16}}$
>
> $p = 10^{20}/n \approx 1.33{\times}10^3$
>
> $N_D \approx n \approx \mathbf{7.5{\times}10^{16}}$

> **(c) $E_F = 0.25$ eV ($< E_i$) ⇒ p-type**
>
> $p = n_i\, e^{(E_i - E_F)/kT} = 10^{10}\, e^{(0.56-0.25)/0.0259} \approx \mathbf{1.58{\times}10^{15}}$
>
> $n = 10^{20}/p \approx 6.34{\times}10^4$
>
> $N_A \approx p \approx \mathbf{1.58{\times}10^{15}}$

> **(d) $N_A = 5\times10^{18}$ cm^{-3} (p-type)**
>
> $p \approx N_A = 5\times10^{18}$
>
> $n = 10^{20}/(5\times10^{18}) \approx 20$
>
> $E_F = E_i - kT \ln(N_A/n_i) = 0.56 - 0.0259 \ln(5\times10^8) \approx$ **0.041 eV**

> **(e) $n = 4.2\times10^{15}$ cm^{-3} (given) ⇒ n-type**
>
> $N_D \approx n =$ **4.2×10^{15}**
>
> $p = 10^{20}/(4.2\times10^{15}) \approx 2.38\times10^4$
>
> $E_F = E_i + kT \ln(n/n_i) = 0.56 + 0.0259 \ln(4.2\times10^5) \approx$ **0.895 eV**

> **(f) minority $n = 100$ cm^{-3} ⇒ strongly p-type**
>
> $p = 10^{20}/100 =$ **1.0×10^{18}**
>
> $N_A \approx p =$ **1.0×10^{18}**
>
> $E_F = E_i - kT \ln(N_A/n_i) = 0.56 - 0.0259 \ln(10^8) \approx$ **0.083 eV**